Automatic Selection of Fine-tuning Distributions for a TabPFN Regressor

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DROPTABLE

Modality 1/3

Motivation

Guiding question: *Given a tabular dataset D, can we try to guess the distribution p(D) it is sampled from?*

The authors of [1] propose three distinct and **fixed** data generators q_{SCM}, q_{BGM}, q_{TA} producing data that looks similar to D. They use this data to fine-tune TabPFN classifiers, with mixed results.

We try to find the best q in the context of a regression task, with the help of AutoML.

[1] M. Bühler, L. Purucker, F. Hutter (2025). Towards Synthetic Data for Fine-tuning Tabular Foundation Models, *Foundation Models for Structured Data workshop at ICML*.

Our Contribution

We define several families of data distributions

$$\{q_\lambda(D) \mid \lambda \in \Lambda\}$$

and use an AutoML pipeline to find the $\lambda \in \Lambda$ for which $q_\lambda(D)$ produces the best fine-tuning data for a TabPFN regressor on a given dataset.

After an initial random sampling phase, we use a multivariate TPE sampler for finding new distributions and prune unpromising runs aggressively. We return the model checkpoint with the highest validation score.

- ☐ Week 1
- ☐ Week 2
- ☐ Week 3
- ☐ Week 4
- ☐ Week 5
- ☐ Week 6
- ☐ Week 7
- ☐ Week 8
- ☐ Week 9
- ☐ Week 10
- ☐ Bonus
- ☐ Literature

Model

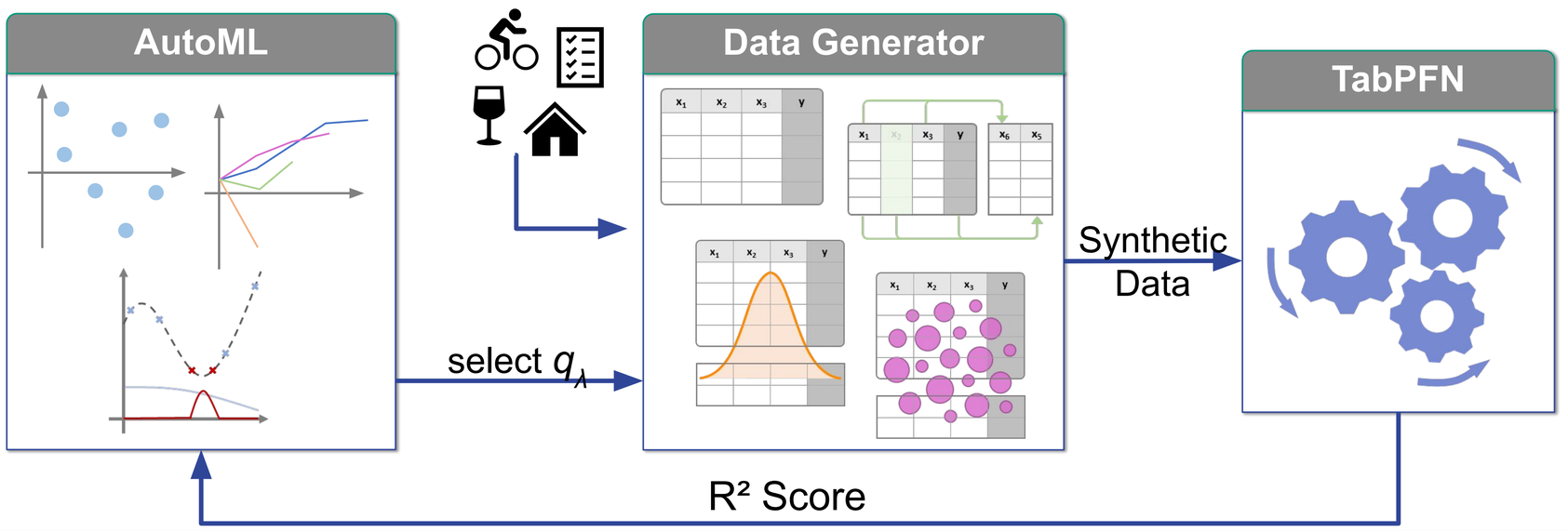


Table Augmentation (TA)
- 16 Parameters

Bayesian Gaussian Mixture (BGM)
- 24 Parameters

Noise
- 9 Parameters

Raw
- 0 Parameters

Resources Used

For development:
- 4 NVIDIA A5000 GPUs
- 2 Intel Xeon 5318Y CPUs
- Total compute estimate: 800 CPU-h

For AutoML:
- Google Colab
- 20h

Workforce:
- 3 full week on average

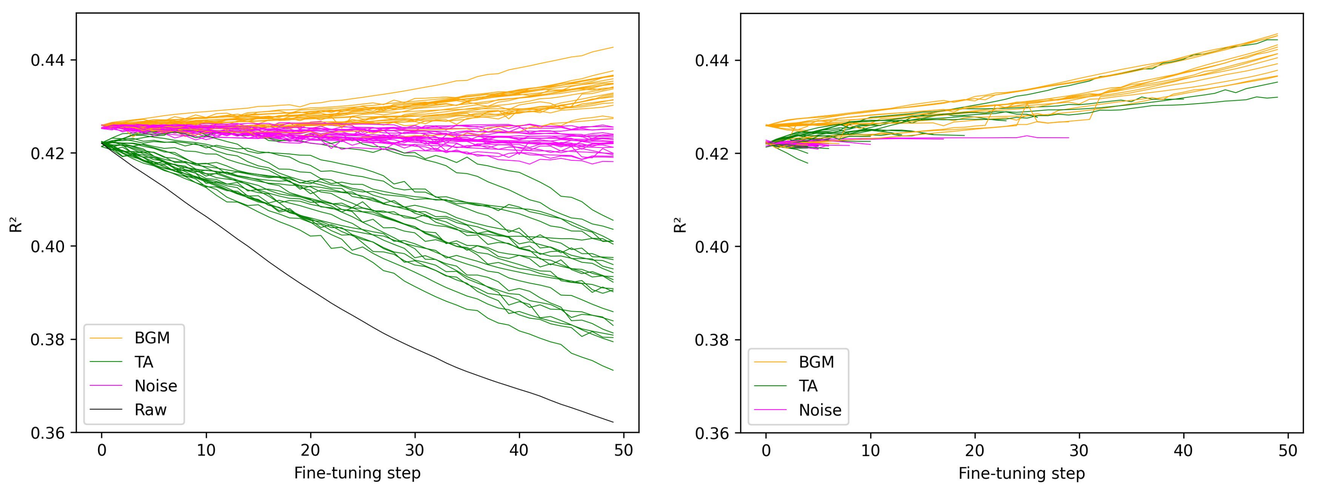
Experiments

We report R²-scores obtained by our fine-tuning procedure on three different datasets. Green entries indicate improvements over the in-context prediction (ICL) of an untuned TabPFN regressor, red entries indicate lower values. Dashes (-) indicate now performance gained by fine-tuning on validation data, and hence no candidate for evaluation of the test data.

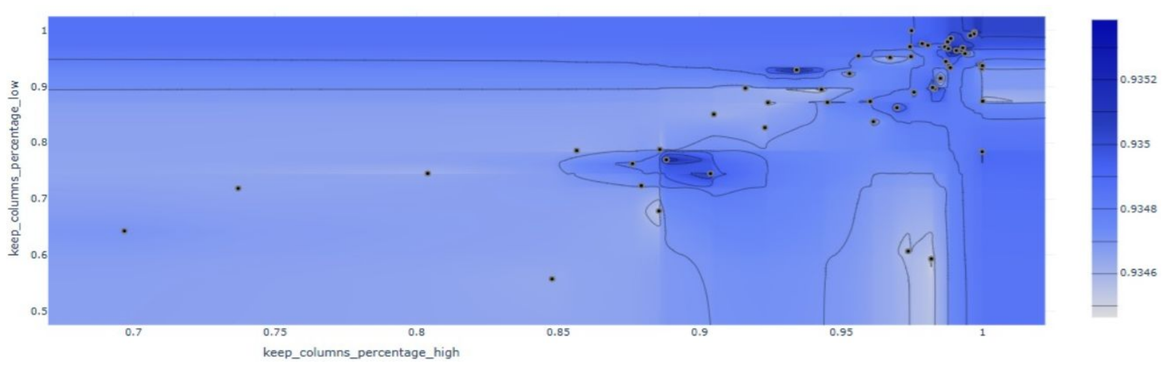
q_λ	ICL	Raw	TA	BGM	Noise
Bike sharing	0.9256	**0.9265**	0.9254	0.9238	0.9263
Brazilian houses	**0.9590**	-	0.9589	-	0.9586
Wine quality	0.4226	-	0.5169	**0.5215**	0.4167

Our final pipeline sequentially performs fine-tuning with all generators and returns the prediction with the highest validation R²:

q_λ	ICL	Raw	TA	BGM	Noise
exam R²-val	0.9088	0.9115	0.9096	0.9103	0.9106
exam R²-test	?	?	?	?	0.9267



Fine-tuning dynamics on the Wine quality dataset. Left: evolution of validation R²-score for 25 trials using a fixed hyperparameter configuration for each of our generator classes; fine-tuning on test data itself in grey. Right: the same, but with distributions produced by our AutoML pipeline.



A contour plot showing the sampling behavior on a high-importance 2-dimensional projection of Λ for the Bike sharing demand dataset, color indicating validation scores.

Number of queries for test score generation: **1**

Discussion

We demonstrate that our AutoML methods can help with the difficult task of fine-tuning a TabPFN regressor (which barely works without an AutoML strategy). But many open directions remain:
- add new and more flexible data generators: SCM based, ensembles, …
- include more aspects of the fine-tuning procedure in the AutoML strategy: learning rate, batch size, …
- with more compute: more datasets, more comprehensive studies, …